

EXPLANATION OF OBJECT-DRIVEN DOM UPDATE PROJECT

In this project, I built a Student Profile Card that demonstrates how JavaScript objects can be used to dynamically update the webpage content (DOM). The main goal was to understand the interaction between JavaScript data (objects) and the HTML elements on the screen.

WORKFLOW:

1. Data Storage using JavaScript Objects:

I created three separate JavaScript objects — `student1`, `student2`, and `student3`. Each object contains all the important information about a student such as:

- `name`
- `age`
- `course`
- `grade`
- `photo` (image URL)

Storing data in objects makes the code clean, organized, and easy to manage. Instead of writing many variables, all related information about one student is kept together in a single object.

2. Connecting JavaScript with HTML (DOM Manipulation):

I gave unique `ids` to the HTML elements (like `studentName`, `studentAge`, `studentCourse`, `studentGrade`, and `studentPhoto`). Then I created a function called `showStudent()` that takes a student number as input.

3. Dynamic Update Logic:

When the user clicks any of the three buttons ("Show Student 1", "Show Student 2", or "Show Student 3"):

- The `showStudent()` function selects the correct student object.
- It then updates the webpage content dynamically using:
 - `.textContent` → to change text inside headings and paragraphs
 - `.src` → to change the student photo

This is called **DOM Manipulation** — changing the HTML content using JavaScript without reloading the page.